

**INTEGRATED CLIMATE-INTELLIGENT SALT
PRODUCTION ECOSYSTEM: A MULTI-MODAL AI
FRAMEWORK FOR SUSTAINABLE YIELD
OPTIMIZATION IN TROPICAL COASTAL
ENVIRONMENTS**

Project ID: 25-26J-431

Project Proposal Report

Perumbuli P G R M D – IT22354310

Supervised by: Dr. Nathali Silva

Co-supervised by: Mrs. Thilini Jayalath

B.Sc. (Hons) Degree in Information Technology, specializing in
software engineering

Department of Software Engineering
Faculty of Computing

Sri Lanka Institute of Information Technology

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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22/08/2025

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22/08/2025

Date:

ABSTRACT

The proposed Federated Learning Waste Valorization framework tackles the urgent need for sustainable waste recovery in Sri Lanka's salt industry while keeping proprietary data safe. Traditional salt processing throws away valuable by-products like bittern and sediments, leading to environmental harm and economic losses. This research introduces Fed Salt, a new federated AI model that optimizes waste valorization across different facilities without sharing sensitive process data. The system uses Differential Privacy techniques to maintain confidentiality while allowing for collaborative improvement.

The methodology combines physics-informed simulations that create synthetic waste datasets with real-world pilot data for validation. The design includes multi-criteria decision-making to balance technical, economic, and environmental factors. A privacy-aware federated digital twin protocol enables decentralized coordination without direct data exchange.

Expected results include better waste recovery yields, less environmental discharge, and higher resource efficiency with strong privacy protections. The research will show how privacy-preserving AI collaboration can speed up innovation in industries that typically work in isolation. It can serve as a model for other mineral processing sectors and guide policy for collaborative sustainability efforts in competitive markets.

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LIST OF ABBREVIATIONS

- AI: Artificial Intelligence
- API: Application Programming Interface
- BOD: Biochemical Oxygen Demand
- CAD: Computer-Aided Design
- CAE: Computer-Aided Engineering
- CLI: Command Line Interface
- COD: Chemical Oxygen Demand
- CPU: Central Processing Unit
- DSS: Decision Support System
- DT: Digital Twin
- EDA: Exploratory Data Analysis
- FL: Federated Learning
- GPU: Graphics Processing Unit
- IoT: Internet of Things
- KPI: Key Performance Indicator
- ML: Machine Learning
- MVP: Minimum Viable Product
- NaCl: Sodium Chloride
- NDAs: Non-Disclosure Agreements
- OT: Operational Technology
- PC: Personal Computer
- SaaS: Software-as-a-Service
- TSS: Total Suspended Solids
- UI: User Interface

- UAT: User Acceptance Testing
- UX: User Experience
- VC: Venture Capital
- XAI: Explainable Artificial Intelligence

1. INTRODUCTION

The Puttalam salterns on the northwestern coast of Sri Lanka are the basis of the solar salt industry in the country, which is powered by large shallow ponds and conventional solar evaporation. These traditional methods, although cheap and eco-friendly when conducted in a country with a stable climate, become highly susceptible to the worsening conditions of climate change. The last few years have produced highly sporadic rain patterns, high and low temperatures, along with unseasonal storms, which have disrupted the evaporation patterns, causing a fall in the predictability of yield, and the self-sufficiency of salt may be in trouble with the nation. The effects of such disruptions travel across the local economy-employment, export potential, and food security. [1]

AI and ML have changed industrial planning and agriculture around the world. Use cases involve predictive climate analytics, computer-vision-based automated quality control, and distributed optimization through federated learning. Interestingly, recently there has been the introduction of AI-enabled platforms, which involve the use of weather information, sensors, and even satellite measurements in order to offer decisive information and dynamic guidance in decision-making both to the salt producers and also to the crop producers. But the effective use of the said technologies in the salt industry, particularly the Puttalam salterns in Sri Lanka, is practically non-existent, thus bringing to light a specific research and innovation gap.

In Puttalam, most of the Sri Lankan salterns still rely on fragmented, empirical management procedures, even though solutions such as Salt Bae demonstrate the efficacy of AI-powered prediction algorithms that smallholder salt-farmers can deploy daily and which leverage multimodal data to drive highly accurate and timely production recommendations. This gap only highlights the need to have a comprehensive decision-support system that integrates planning, forecasting and quality control into a coherent, practical model.

There has been a tendency historically to focus the interventions of salt production on specific improvements, such as those of mechanisation or filtration, without considering adequately environmental or operational data, as well as market data. The suggested Integrated Climate-Intelligent Salt Production Ecosystem is thus a breakthrough since it uses an integrated digital platform with advanced environmental forecasting algorithms, adaptive planning models, real-time quality control overseen by computer vision, and federated learning empowered waste valorisation.

In this ecosystem, the Adaptive Intelligent Planning Engine plays the critical role relating predictive analytics to flexibility in operational performance. It dynamically combines instantaneous market information, regulations, and user preferences to feed resilience and profitability in the climate-sensitive environment of Puttalam. The project hopes to enhance the resilience, efficiency, and sustainability of the most significant salt-producing area situated in Sri Lanka because the project will modify the manner of salt production in the country, which is currently reactive and stuck in tradition to a pro-active, AI-based concept, which will ensure the lasting social and economic, and environmental gains.

1.1. BACKGROUND AND LITERATURE SURVEY

One of the most ancient industrial processes is the extraction of salt using the method of solar evaporation, which is traditionally observed in the tropical coastal settings, since high temperatures and scarcity of rainfall establish an environment in which salt can easily crystallize (Jones et al., 2018). [2] Sri Lanka has major salterns in Hambantota, Puttalam, and Elephant Pass that plays a significant role in supplying salt in the country (Perera & Silva, 2022). [3] However, this has changed drastically over the past few years because of seasonal anomalies and heightened climate variability, which has complicated the conventional production cycles making them adapt to the operations without the use of effective predictive and planning tools (Fernando et al., 2021). [4]

Present operational planning in these salterns is relying mostly on manual estimation and past averages usually become inadequate in unpredictable weather conditions. The absence of integrated decision-support systems makes operators provide crucial decisions based on incomplete information (weather predictions of publicly funded weather agencies, market prices provided by traders, and informal estimates of yield using subjective judgment of experience) leading to inefficiencies and lost reliability in production (Wijesekara & Rajapakse, 2020) [5].

The use of AI in production planning has proved to be successful in other countries that are in close reliefs in terms of resource quest. As such, in agriculture, AI can be used to plan crops adaptively based on weather, soil, and markets data to maximise yields (Zhang et al., 2020) [6] . Predictive analytics are used in the fisheries industry, in which it runs on maximizing fish quotas to operate in an unstable environment (Duarte et al., 2019). [7] Likewise, in manufacturing, the reinforcement learning algorithm is utilized in improving resource scheduling and operation prowess (Wang & Zhang, 2021). [8] Such global progress notwithstanding, these AI-driven methodologies have not effectively been modified to deal with the unique interaction of the environmental, economic, and regulatory complexities that exist in the salt industry in developing nations such as Sri Lanka.

Sri Lanka has a salt industry facing two kinds of pressure which are imposed on it by the market dynamics and by the regulatory frameworks. During domestic trading, the spheres need stable plans and cheap salt, and in communities abroad, traders must meet strict purity standards by international agencies (FAO, 2022) [9]. Respectable operational planning system has to wade through these perhaps competing objectives without compromising on economy as well as respecting environmental sustainability requirements (Ministry of Trade, Sri Lanka, 2023). [10]

In this literature, there is an urgency to use integrated data-driven decision-support solutions involving climatic forecasts and market trends, as well as regulatory requirements that will facilitate the more proactive and efficient operational planning

in Sri Lankan salterns, and most importantly in Puttalam. This growth of such systems holds out the promise of enhancing adaptation to climate variability, maximization of resource utilization as well as quality of products and hence will tackle problems around the globe and locally affecting the industry.

1.2. RESEARCH GAP

It has been seen that even though many countries in the world have developed and implemented AI-driven operational planning within an industry like agriculture, fisheries, and manufacturing, an industry-specific and integrated system with regards to salt production is not found in Sri Lanka. Current digital applications in adjacent industries are concerned either with simple weather notification or narrow areas of operation, rather than building a more complete, multi-parameter decision-making regime, specific to salt production.

The key research gaps identified and addressed by the components of the proposed Integrated Climate-Intelligent Salt Production Ecosystem are:

- **The absence of Multi-Parameter Integration:**
Current approaches to planning in the salt industry do not integrate its main inputs such as climate predictions, changing production capabilities, demand levels, operational expenses, and legal requirements. This shortcoming is filled by Component 1: Crystallization Forecasting, which runs on multimodal data crystallization forecasting that incorporates weather forecasts and IoT sensor measurements and takes the form of highly accurate data-driven forecasts of optimal crystallization conditions and crop harvesting time.
- **Reactive, as opposed to Predictive Decision-Making:**
Existing methods only make decisions based on the observed environmental changes and not on the ability to predict the changes, which results in a loss of efficiency. This gap is filled by component 2, the Adaptive Intelligent Planning Engine (Intelligent Planning Engine) that combines forecasts, market trends, cost structures, and regulatory requirements into frame-based predictive operational plans that make proactive management proactive.
- **Failure to Adapt in Real Time to New Conditions:**
The market conditions and salt production environments are very unstable and quick to change with regard to weather and demand. Nonetheless, existing systems do not have the capability of automatically varying plans on an almost real-time basis. This obstacle is circumvented by Component 3, which is Computer Vision Quality Control that facilitates real-time automated monitoring of salt quality

feeding real-time feedback into operational changes, and Component 2, continual refreshes of planning strategies using real-time input data.

- Lack of planning tools relevant to the stakeholders:

The business strategy of saltern operators differs; the owners of one saltern are oriented to the production of high-quality salt that covers export needs whereas others want to increase their output volume to sell it into local markets. Nowadays, there are no tools with such possibility. The user input and customization layer (Component 2) builds into the Adaptive Intelligent Planning Engine a customized level of operation strategies in line with the specific priorities of individual stakeholders.

- AI Techniques That Are Not Utilized to the Full Potential in Salt Production Intrigues:

Though AI technologies like reinforcement learning, fuzzy logic, and heuristic optimization are well-developed in other domains, they are not applied to the conditions and the possibilities of salt production in the tropics. The system, where all components are integrated (namely, Component 4: Federated Learning Waste Valorisation), utilizes state-of-practice AI solutions, including federated learning algorithms, to make it possible to coordinate the optimization processes across numerous locations, all the time protecting the privacy of data.

Coordinating these interconnected gaps through a modular yet integrated ecosystem, this project aims at changing how the Sri Lankan salt industry, specifically the Puttalam salterns, is currently operated, due to the current fragmented and reactive management it is based upon, towards sustainable, data-driven, and resilient operational planning.

1.3. RESEARCH PROBLEM

The key research question that will prove to be the impetus of this venture can be put in the following words:

“How would an adaptive, learnt planning algorithm be formulated to incorporate real time environmental, production, economic factors, and regulatory factors, in the context of generating optimal operational strategies to enhance profitability, efficiencies, and make the salt production industry of Sri Lanka import-dependent with regards to climatic variability?”

This problem statement captures both the technical challenge - creating and deploying an effective, multi-parameter optimization tool that can address complex and dynamic inputs and the practical need, giving salt producers access to sophisticated decision-making tools to respond to an increasingly erratic climate and market environment.

The issue can even be further broken down into three critical dimensions:

- **Environmental Uncertainty:** Dealing with unpredictable changes in weather and climate conditions that affect the salt crystallization process and harvest directly and are largely dependent on changes in temperature, rainfall, and other weather-related conditions.
- **Economic Variability:** Adjusting the production schedule and resource allocation to varying and changing market demand, availability, and price signals in a way that can maximize revenue and minimize wastage and overproduction.
- **Regulatory Compliance:** This is necessary to ensure that operational plans and final products are acceptable as per the strict national and international quality standards and environmental regulations, which are vital in sustaining export market capability and industry sustainability practices.

Solving this complex issue will not only increase the level of self-reliance of salt production in Sri Lanka but also help in increasing the competitive ability of the industry in the local and international markets. Moreover, the plan of developing the solution as a scalable and modular framework will facilitate transferability of the project to other tropical coastal areas of the world, which might be similar to those of the project areas in terms of environmental and operational similarities in salt production.

Industrial salt waste valorization processes, essential for environmental sustainability and resource recovery, are currently hampered by data silos, privacy concerns, and the inherent complexity of optimizing operations across decentralized industrial environments. Existing centralized AI and decision support systems are inadequate due to their inability to securely and collaboratively leverage distributed data, leading to sub-optimal resource recovery, increased operational costs, and persistent

environmental risks. The central research problem is to simultaneously overcome the challenges of data privacy, decentralization, and operational complexity to achieve truly collaborative and optimized industrial salt waste valorization. This involves designing a framework that can learn from distributed, sensitive industrial data, simulate complex valorization processes in real-time, and provide actionable, privacy-preserving recommendations to human operators across a network of disparate facilities. The problem is not merely technical; it is an interconnected challenge where optimizing one aspect (e.g., efficiency) often compromises another (e.g., privacy), forming a complex conundrum. Without a framework that simultaneously addresses privacy, decentralization, and optimization, industries will continue to face significant barriers to achieving sustainable and economically viable salt waste management.

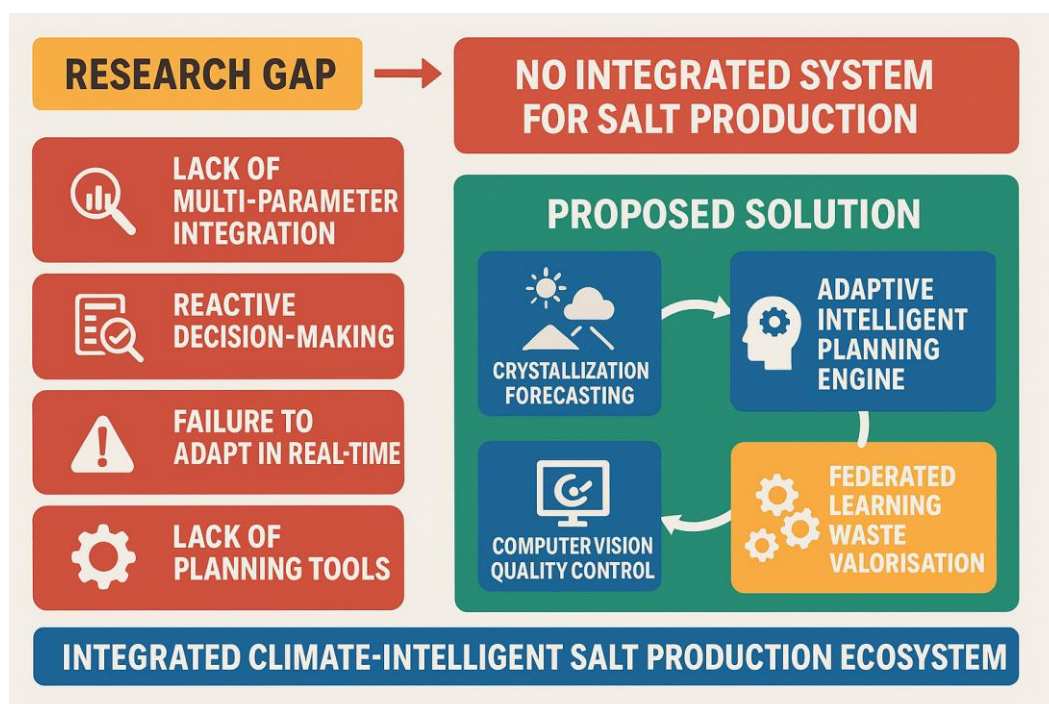


Figure 1- Research Gap

2. RESEARCH OBJECTIVES

This section introduces the overall aims of the Integrated Climate-Intelligent Salt Production Ecosystem as a multi-component platform to modernize, optimize and future-proof climate-sensitive coastal salt production in Sri Lanka.

2.1. MAIN OBJECTIVE

In producing and deploying a bespoke, AI-powered ecosystem designed to support the salt production industry in Sri Lanka by facilitating predictive environment analytics, adaptive operating program code, autonomous quality control, and joint optimization of operations resources, which works toward increased productivity, sustainability, and climate adaptation, along the entire salt production value chain.

SMART Breakdown:

- **Specific:** Design and implement an integrated platform that would consist of four autonomous but interconnected AI-powered modules (crystallization forecasting, intelligent planning, computer vision quality control, federated learning waste valorization).
- **Measurable:** The performance of the success is measured through increases in production yield ($\geq X\%$), use efficiency in resources ($\geq Y\%$), quality product compliance ($\geq Z\%$), and the decrease of wastes as well as positive feedback on at least two operational pilots.
- **Achievable:** Achievable by leveraging the project team's expertise in AI, software engineering, and salt industry operations, and by collaborating with local saltern partners for piloting and validation.
- **Realistic:** Realistic to the available technology and ability to access experienced personnel in operation, besides high stakeholder involvement in the industry.
- **Time-bound:** To get fully implemented, deployed, and reviewed even on a project level.

2.2. SPECIFIC OBJECTIVES

The specific objectives below specify the nature and extent of the contributions of each ecosystem component as well as promoting synergy among the components:

- **Crystallization Forecasting:** To develop and validate machine learning models of weather, sensor, and historical production data, and give early and precise deep forecasts of optimal crystallization and harvest windows.
- **Intelligent Planning Engine:** Develop flexible AI-based planning systems that can take predictive forecasts, real-time market, manufacturing, and regulatory information, and convert it into stakeholder-optimised, actionable operational plans to make profitable, pro-active decisions.

- **Computer Vision Quality Control:** Implement a high-throughput automated computer vision system to grade the salt and detect salt impurities in real-time so that the high-quality and regulatory standards of the product can be achieved, and any human touch and production bottlenecks are reduced.
- **Federated Learning Waste Valorization:** Develop a federated learning infrastructure, a secure and collaborative setting to optimize the reuse and valorization of salt waste in a distributed set of sites, without compromising data that is sensitive but also enhancing a sustainable circular and economic enrichment initiative.

In attaining these goals, the project will not only assist Sri Lanka in the self-sufficiency in salt production, competitiveness in both regional and international markets, and sustainable and technology-driven management of its resources, but also helping provide the country with the necessary leadership in the above aspects. The replicability of the ecosystem in terms of modularity and scalability also allows the ecosystem to be implemented in other tropical coastal nations that are grappling with the same challenges and thereby making more contributions to the topic of climate-smart industry transformation.

3. METHODOLOGY

3.1. System Overview

The Integrated Climate-Intelligent Salt Production Ecosystem is a modular, AI-driven platform (potentially modular) which aims to revolutionize the salt production industry in Sri Lanka by performing end-to-end decision support tasks to the entire salt production lifecycle. The structure of its architecture is divided into four related components that take care of each of the vital steps in operational planning, execution, and optimization.

System Components

1. **Crystallization Forecasting:** Combines multimodal information (weather, IoT sensor networks, and satellite imagery in addition to historical production information) with advanced time series models (ARIMA, Prophet, and LSTM) to determine the ideal conditions to crystallize salt and harvest it. This makes it possible to develop proactive plans and to reduce the risks of yields depending on the uncertainty of the climate.
2. **Intelligent Planning Algorithm:** Converts predictive intelligence of Component 1, with real-time market trends, production capacities, economic data and regulatory requirements into optimized operating strategies implemented into actions. It uses multi-parameter optimization, heuristics, fuzzy logic, decision trees, and reinforcement learning to support dynamic data-driven planning which can change as circumstances change.
3. **Quality Control: Computer Vision:** Automatically grades the salt quality using computer vision image analysis and impurities are detected in real time. This improves management without the need of manual check-up thereby making it more regulated and assuring the quality of the products to both the local and international markets.
4. **Federated Learning Waste Valorization :** Facilitates the distributed and privacy-preserving coordination of different sites of the saltern to optimize by-product reuse and valorization. By using federated learning and differential privacy, they can be sure that critical operational data will not be shared directly, but it will lead to innovation and data safety .

Key Features

- Modularity
- Real-time Data
- User-Centric Dashboard

Such an interconnected ecosystem has the unique opportunity to consider the complex climate sensitive operation realities of the salt industry in Sri Lanka.

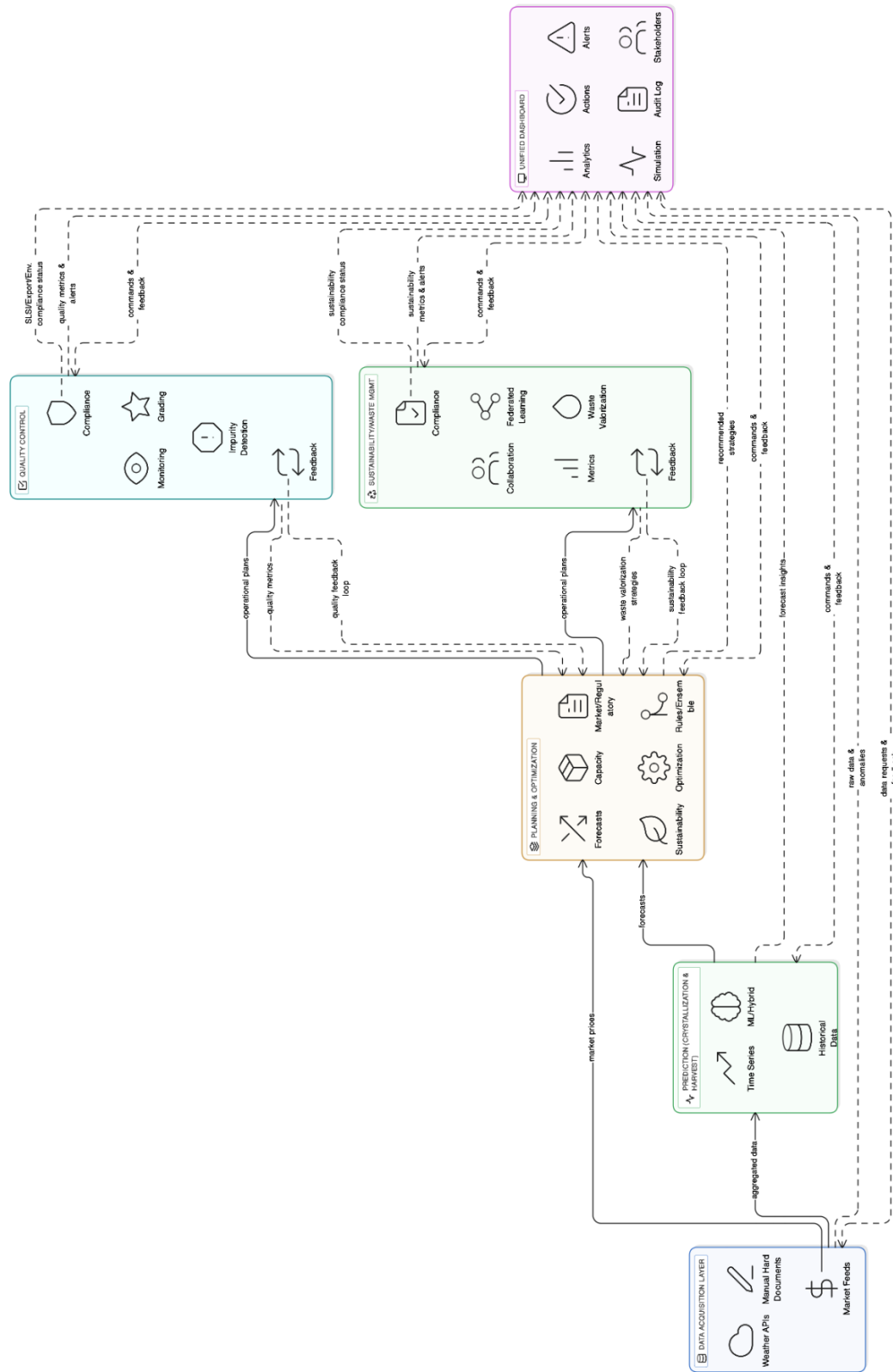


Figure 2 – System Diagram

3.2. Individual Component

3.2.1. Requirement Gathering and System Design

The proposed system integrates multiple AI components to support sustainable and intelligent salt production. To ensure its effectiveness, requirement gathering was carried out across three main dimensions: stakeholder needs, technical requirements, and data requirements.

Stakeholder Needs

- Saltern owners require a method to extract value from waste streams (bittern, sediments, and brine residues) while maintaining confidentiality of proprietary processes.
- Competing industrial players (food, chemical, desalination sectors) need a secure mechanism to collaborate without exposing sensitive operational data.
- Environmental regulators demand sustainable waste management strategies that reduce ecological impact.
- Decision-makers need tools to evaluate trade-offs between technical yield, cost-effectiveness, and environmental sustainability.

Technical Requirements

- Federated Learning Framework: Must allow distributed model training across multiple facilities without centralizing raw data.
- Differential Privacy Integration: Protects model updates so no sensitive process details can be reconstructed.
- Synthetic Data Support: Ability to generate physics-informed synthetic salt waste datasets for training and validation when real data is scarce.
- Heterogeneous Data Handling: Support for non-IID data across facilities, with personalization or client clustering.
- Digital Twin Integration: A federated digital twin protocol for simulating and coordinating distributed waste recovery processes.
- Decision Support System: Multi-Criteria Decision Making (MCDM) to balance environmental, economic, and technical objectives.

Data Requirements

- Industrial Process Data: Waste composition, recovery rates (NaCl, Mg, K, Cl), and energy consumption metrics.

- Synthetic Data: Simulated crystallization and concentration dynamics to augment limited real-world data.
- Operational Variability Data: Sensor inaccuracies, thermal dynamics, and brine concentration fluctuations.
- Market Data: Demand and price trends for by-products (e.g., magnesium salts, chlorine).

3.2.2. Development of the Web Platform

The development of the user-facing web platform will adhere to a modular, microservices-based architecture to ensure flexibility, scalability, and ease of maintenance. This approach will allow for independent development, deployment, and scaling of different functionalities. The core of the platform will leverage established open-source frameworks for FL, such as TensorFlow Federated and Flower, and for DT, utilizing environments like Unity for 3D modeling and integrating with CAD/CAE tools for accurate digital representations. Building such a complex integrated system from scratch would be prohibitively time-consuming and resource-intensive. Leveraging mature open-source tools not only accelerates development but also enhances interoperability and benefits from extensive community support. This strategic choice allows the project to focus its novel contributions on the intricate integration logic, the development of domain-specific AI models for salt waste valorization, and the design of the decentralized DSS components, rather than re-inventing foundational AI/DT infrastructure.

A critical aspect of the web platform will be the design of its User Interface (UI) and User Experience (UX) for the Decision Support System. Intuitive dashboards and visualization tools will be developed to present complex data and AI-driven insights in an easily digestible format for diverse users, including plant operators and management personnel. These visualizations will encompass process parameters, predicted anomalies, optimization recommendations, and resource recovery metrics. The platform will also integrate interactive "what-if" simulation capabilities, directly accessible through the UI, allowing users to test different scenarios and understand potential outcomes before implementing changes in the physical valorization process. Furthermore, explainable AI (XAI) components will be integrated to provide transparency into the AI model's reasoning and the DSS recommendations, fostering trust and enabling informed human oversight. Simply displaying data or predictions is insufficient for an effective DSS; the system must provide actionable insights and facilitate human-AI collaboration. Poor UI/UX, a lack of explainability, or an inability to interact with the models would lead to underutilization or mistrust, negating the system's potential benefits. Therefore, this development phase will focus on transforming raw data and complex model outputs into meaningful, context-aware, and trustworthy recommendations that empower human decision-makers, recognizing that the success of the framework relies heavily on its usability and the ability of human operators to understand and act upon its intelligence.

3.2.3. Testing and Quality Assurance

A comprehensive, multi-stage testing approach will be implemented to ensure the reliability, accuracy, security, and performance of the integrated framework. This rigorous process is essential for building a trustworthy system in a high-stakes industrial environment.

- Unit Testing will verify that individual components, such as FL algorithms, DT models, and DSS modules, function correctly in isolation.
- Integration Testing will ensure seamless data flow and interaction between the FL, DT, and DSS components, validating their interoperability.
- System Testing will validate the end-to-end functionality of the entire framework within simulated decentralized industrial environments, mimicking real-world operational conditions.
- Performance Testing will rigorously evaluate the system's scalability, latency, and resource consumption under varying data loads and client numbers. This will ensure the system can handle the demands of large-scale industrial deployment.
- Security Testing will involve proactive penetration testing and vulnerability assessments to identify and mitigate potential threats, including poisoning attacks, inference attacks, and data breaches, which are significant concerns in federated and decentralized systems.
- Privacy Compliance Audits will verify that privacy-preserving mechanisms, such as differential privacy, homomorphic encryption, and secure multi-party computation, are correctly implemented and meet predefined privacy guarantees and regulatory standards.

User Acceptance Testing (UAT) will involve direct participation from industrial stakeholders to validate the system's usability, relevance, and effectiveness in supporting their day-to-day operational needs.

3.2.4. Data Collection and Analysis

The success of the **FedSalt framework** relies on robust, high-quality data representing the key aspects of the salt waste recovery process. For a final-year project, the data can be a combination of **synthetic, simulated, and small-scale real measurements**. This ensures feasibility while capturing the essential features needed for federated learning and decision support.

Key Data Types

1. Raw Waste Data

- Focus on **major salts and ions**: NaCl, Mg, K, Cl.
- Optional: small number of trace components (simulated heavy metals, TSS, turbidity).
- Synthetic values can be generated using **random distributions or simple chemical rules** to reflect typical saltern waste streams.

2. Operational Data

- Simulate basic **process parameters**: temperature, flow rates, evaporation time, energy usage.
- Use controlled ranges to represent site variability (e.g., Site A is hotter than Site B).

3. Output Data

- Yields and purity of recovered salts (NaCl, Mg salts) can be **computed from input + process parameters** using simple formulas.
- Optional: add small random noise to simulate real measurement uncertainty.

4. Equipment Data

- For MVP, track only **basic operational health metrics**: motor current, vibration level, or a simple “status OK / Maintenance Required” label.
- This can be **synthetic or small-scale experimental data**.

5. Environmental Data

- Include **temperature, humidity, solar radiation** as simple numeric features.

- Use historical weather data or generate **synthetic daily variations** for each site.

Data Collection Methods

1. Synthetic Data Generation

- Use **Python + NumPy/Pandas** to generate datasets for multiple “sites.”
- Ensure **non-IID distributions** to simulate realistic heterogeneity between facilities.
- Example: Site A has higher Mg concentration; Site B has lower evaporation rates.

2. Small-scale Lab or Field Measurements

- If access is available, collect **a few samples** of brine or salt waste in a controlled lab setting.
- Measure NaCl, Mg, and K concentrations using simple titration or conductivity methods.

3. Simulated Historical Data

- Generate sequences representing **operational history**, equipment usage, and output yields over time.
- Use this data to test model training, federated aggregation, and decision support logic.

Data Preprocessing and Analysis

- **Cleaning & Validation:** Ensure synthetic or collected data is within realistic ranges.
- **Normalization & Encoding:** Scale numeric features; one-hot encode categorical variables (e.g., equipment status, stream type).
- **Exploratory Data Analysis (EDA):**
 - Visualize distributions for each site.
 - Compare variability between sites to highlight non-IID behavior.
 - Identify correlations between operational parameters and output yields.

- **Partitioning for FL:** Split each site's dataset into **train, validation, and test sets**, maintaining realistic variability.

3.3. Materials Required to Carry Out the Project

3.3.1. Equipment

Development Workstations: High-performance desktops or laptops for model development, training, and testing. Requirements:

- CPU: $\geq$ Intel i5/i7 or AMD Ryzen 5/7
- RAM: $\geq$ 16 GB
- GPU: Optional for faster training (e.g., NVIDIA GTX 1660 or higher)
- OS: Windows/Linux/macOS

IoT Devices / Sensors (Simulated or Real):

- To collect environmental and operational data: temperature, humidity, solar radiation, flow rates, and pH.
- For MVP, these can be simulated using Python scripts generating time-series data.

Smartphones / Tablets: For dashboard testing, alerts, and operator interface demonstration.

Servers / Cloud Infrastructure:

- Local: Use Docker containers to simulate multiple sites and a central server.
- Cloud (optional): AWS EC2, GCP, or Heroku to host the FL aggregator, dashboard, and decision-support APIs.

Laboratory Equipment (Optional): Basic lab tools (pH meter, conductivity meter) if collecting small-scale real data; otherwise, data can be fully synthetic.

3.3.2. Software and Computer Support

Programming Languages & IDEs:

- Python 3.x (for federated learning, ML models, and simulation scripts)
- JavaScript/TypeScript + Node.js (backend APIs)
- React (frontend dashboard)

- IDEs: Visual Studio Code, PyCharm

Version Control: Git + GitHub for collaboration and backup

Federated Learning Framework:

- **Flower (FL)** for client-server orchestration
- PyTorch or TensorFlow for model definitions
- Opacus (PyTorch) or TF Privacy for differential privacy integration

Web Frameworks & APIs:

- Backend: FastAPI or Express.js for REST APIs and decision support endpoints
- Frontend: React + Chart.js/Plotly for dashboard visualizations

Database Systems:

- Local client storage: SQLite or JSON files for site-level datasets
- Central metadata / logging: PostgreSQL or MongoDB

3.3.3. Graphics

- **UI/UX Design Software:** Figma for designing intuitive dashboards and operator interfaces.
- **Visualization Libraries:** Matplotlib, Seaborn, and Plotly for rendering real-time process insights and decision-support outputs.
- **Multimedia Creation Software:** Adobe Illustrator, Canva for diagrams, flowcharts, and presentation graphics used in system documentation.

3.3.4. Other Resources

- **Expert Domains:** Collaboration with chemical engineers, salt industry specialists, and AI researchers to validate waste valorization models.
- **Third-Party APIs:** Integration of APIs for weather/climate data, market intelligence, and secure federated learning protocols.
- **Task Management Tools:** Jira, Trello, or Asana for project planning, team coordination, and milestone tracking.
- **Datasets:** Historical production logs, laboratory analysis records, synthetic datasets, and real-time IoT sensor data for training and testing models.

Implementation Notes for Feasibility

- All **FL clients and server** can be run locally using **Docker containers** to simulate multiple sites.
- Real sensor data is optional; synthetic data can fully demonstrate the system workflow.
- Differential privacy and secure aggregation can be implemented using **Opacus + Flower custom strategy**.
- The **dashboard** can consume real-time updates via WebSockets or REST APIs from the central server.
- Decision-support module (MCDM/optimizer) can use **Python libraries** (e.g., PuLP or scikit-criteria) for route planning and trade-off evaluation.

3.4. Data Required for the Project and Collection Methods

Raw Salt Waste Characterization: This will include detailed chemical composition data, such as concentrations of key salts (e.g., NaCl, Na₂SO₄, KCl), heavy metals, and other inorganic impurities. Crucially, the types and concentrations of organic matter (e.g., COD, BOD, TOC, and specific organic compounds like dicamba, dichlorophenol, ortho-cresol, which are challenging to treat) will be collected. Physical properties such as pH, conductivity, density, turbidity, and total suspended solids (TSS) will also be essential.

Process Operational Data: This encompasses continuous measurements of temperature and pressure at various reaction vessels, evaporators, and crystallizers. Flow rates and levels in tanks, pipelines, and input/output streams will be monitored. Data on chemical dosing, including the amounts and types of reagents added (e.g., oxidizers, precipitants), will be critical for process control and optimization. Energy consumption (electricity usage for pumps, heaters, aeration blowers, and overall heat consumption) will be tracked to assess efficiency. Reaction times for various treatment steps will also be recorded to understand kinetics and optimize throughput.

Output Product/Effluent Data: This category includes the purity and yield of recovered salts and minerals (e.g., NaCl, Na₂SO₄, K₂SO₄), and the yield of other valuable minerals that can be extracted from brine (e.g., lithium, magnesium, potassium). Treated water quality will be assessed against environmental compliance parameters such as COD, BOD₅, NH₄-N, N_{tot}, P_{tot}, pH, and conductivity.

Equipment Health Data: This involves collecting data on equipment vibration, motor current, bearing temperature, historical maintenance logs, and anomaly alerts, which are vital inputs for predictive maintenance models within the Digital Twins.

Environmental Data: In line with the proposal's broader scope, local climate conditions (temperature, humidity, solar radiation) will be considered, especially if the valorization process incorporates solar evaporation or other climate-dependent methods.

Historical Data Integration: Existing historical operational logs, maintenance records, and laboratory results will be collected from participating facilities. This historical data is crucial for building initial models, providing a baseline for comparisons, and enabling long-term trend analysis and optimization.

3.5. Estimated Time Frame

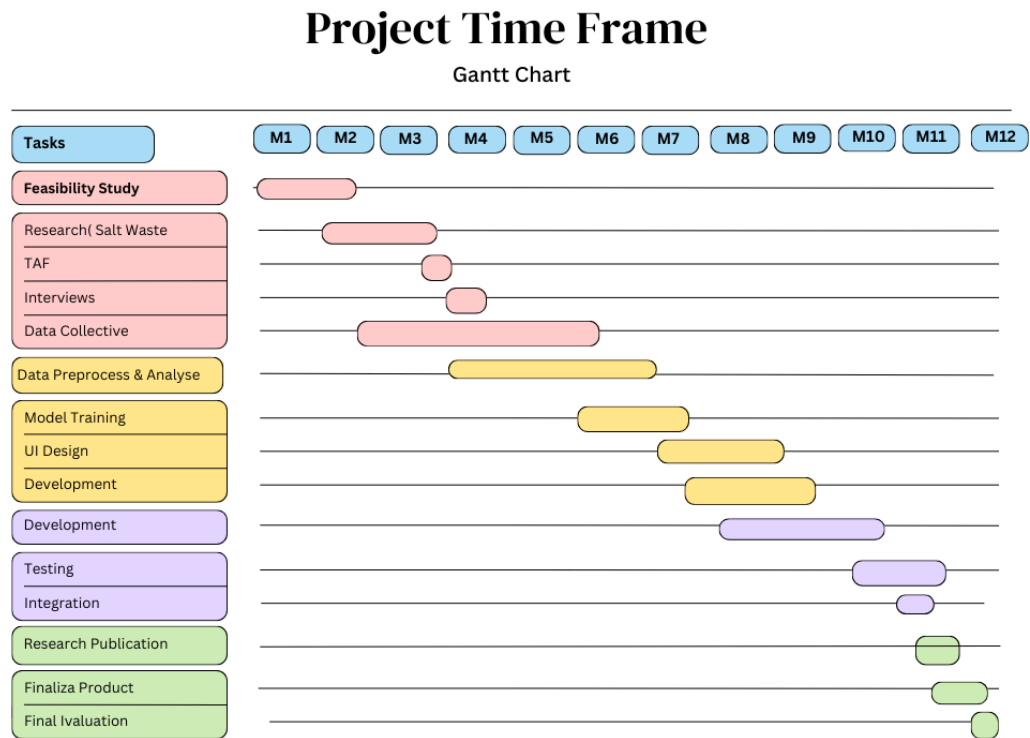


Figure 3 – Time Frame

3.6. Commercialization

3.6.1. Product Differentiation and Value Proposition

The framework's core differentiation lies in its unique ability to enable collaborative, privacy-preserving optimization of salt waste valorization processes. This addresses a critical need in decentralized industrial environments that cannot be met by traditional centralized solutions, which often face insurmountable barriers due to data privacy and intellectual property concerns. The proposed framework directly addresses this latent market need for collaborative intelligence without compromising sensitive information. The value proposition extends beyond mere technical efficiency; it solves a fundamental trust and data governance problem that unlocks new levels of inter-organizational collaboration for waste valorization.

Key value propositions include:

- **Enhanced Resource Recovery:** Maximizing the extraction of valuable minerals and water from waste brine, contributing directly to a circular economy.
- **Reduced Environmental Impact & Compliance:** Minimizing hazardous waste disposal and ensuring adherence to increasingly stringent environmental regulations.
- **Operational Cost Reduction:** Optimizing energy and chemical consumption, implementing predictive maintenance to reduce downtime, and enabling efficient process control.
- **Data Privacy & Security:** Enabling collaboration without compromising sensitive proprietary data or intellectual property, a critical concern for industrial clients.
- **Decentralized Control & Scalability:** The framework is inherently suited for distributed industrial operations, offering flexibility, resilience, and easy onboarding of new sites.
- **Improved Decision-Making:** Providing real-time, AI-driven insights and "what-if" simulations for proactive process management, empowering operators with actionable intelligence.

3.6.2. Revenue Model

- The primary revenue model will be a **Subscription-based Software-as-a-Service (SaaS)** offering. This will involve tiered subscriptions, potentially based on factors such as the number of connected facilities, data volume processed, or the level of advanced optimization features accessed. This model aligns with the industry trend where manufacturing companies are increasingly seeking solutions to reduce operational costs through smart technologies.
- In addition to SaaS, **Value-added Services** will be offered, including consulting for initial implementation, custom model development tailored to specific industrial processes, and ongoing maintenance and support. A hybrid
- **Performance-based Pricing** model could also be explored, where a portion of the revenue is tied directly to the quantifiable savings or resource recovery improvements achieved by the client. This approach would de-risk client investment and accelerate market penetration by directly aligning the revenue model with the value delivered. For highly sensitive industrial environments requiring on-premise solutions, a **Licensing** model for the core software components could be considered. The commercial strategy will emphasize the quantifiable economic benefits of the framework (e.g., X% reduction in waste, Y% savings in energy) to industrial clients, moving beyond a purely feature-based sales approach.

3.6.3. Scalability and Market Penetration

- The framework's inherent design supports robust scalability and efficient market penetration. The decentralized nature of Federated Learning allows for the easy onboarding of new industrial sites without compromising privacy or requiring massive central infrastructure upgrades. This horizontal scalability is a significant advantage for expanding the solution across a diverse industrial landscape. Furthermore, the modular deployment of components will allow clients to start with core functionalities and expand their adoption as their needs evolve.
- Initial market penetration will focus on industries generating significant salt waste, such as chemical manufacturing (e.g., chlor-alkali plants), desalination facilities, and mining operations. Strategic partnerships with industrial automation providers, environmental technology firms, and relevant industry associations will be pursued to leverage existing networks and accelerate adoption. The FL component thrives on more participants, as the global AI model becomes more robust and accurate with increased data diversity, creating a positive feedback loop or network effect. This network effect makes the solution more attractive to new participants, accelerating market penetration. The commercial strategy will actively promote this collaborative benefit, positioning the framework not just as a tool, but as a shared intelligence ecosystem for the salt waste valorization sector.

3.6.4. Marketing Strategy

The marketing strategy will focus on effectively communicating the framework's advanced technical capabilities in terms of clear, quantifiable business and environmental benefits to industrial decision-makers. This involves translating complex AI, Digital Twin, and Federated Learning innovations into compelling value propositions, such as "reduce operational costs by X%," "achieve Y% higher resource recovery," and "ensure data privacy compliance".

Key marketing activities will include:

- **Content Marketing:** Developing detailed case studies, white papers, and webinars that showcase the framework's success in pilot deployments, emphasizing the quantifiable benefits achieved in terms of cost savings, environmental impact reduction, and privacy protection.
- **Industry Conferences & Trade Shows:** Presenting research findings and demonstrating the platform at leading industrial, chemical engineering, and environmental technology events to engage directly with potential clients and partners.
- **Direct Sales & Account-Based Marketing:** Implementing targeted sales strategies to engage with large industrial players and conglomerates that face significant salt waste challenges.
- **Thought Leadership:** Publishing articles in reputable industry journals and participating in expert panels to establish the research team and the framework as authoritative voices in sustainable industrial AI.

3.6.5. Support and Maintenance

- **Tiered Support:** Offering various levels of technical support (e.g., 24/7, business hours, dedicated account managers) based on client subscription tiers to cater to diverse operational needs.
- **Regular Updates & Patches:** Providing continuous software updates, security patches, and feature enhancements to ensure the system remains current, secure, and performant.
- **Remote Monitoring & Diagnostics:** Utilizing the framework's own capabilities, such as Digital Twins for system health monitoring, to proactively identify and resolve issues before they impact operations.

- **Training & Documentation:** Offering comprehensive training programs and user documentation for operators and IT staff at client sites to ensure effective adoption and utilization of the system

3.6.6. Intellectual Property Protection

- **Trade Secrets:** Protecting proprietary datasets (if any are developed for internal testing), specific model architecture, and detailed implementation know-how as trade secrets.
- **Copyright:** Copyrighting the software code, documentation, and user interface designs.
- **Licensing Agreements:** Implementing robust software licensing agreements that clearly define usage terms, restrictions, and intellectual property ownership for commercial deployments.
- **Non-Disclosure Agreements (NDAs):** Requiring all collaborators, employees, and partners involved in development and deployment to sign NDAs to protect sensitive information.
- **Ethical Data Sourcing:** Ensuring that all data used for training (even if federated) is sourced ethically and legally, with proper permissions, to mitigate potential IP infringement claims.

3.6.7. Funding and Investment Opportunities

- To ensure the successful development and commercialization of this innovative framework, a diversified funding and investment strategy will be pursued. Initial research and development will be supported through **Research Grants** from public and private funding bodies, including environmental agencies, industrial innovation funds, and national science foundations.
- For scaling the solution and market entry, **Venture Capital (VC) and Angel Investment** will be sought from investors specializing in Industry 4.0, cleantech, environmental technologies, and AI/IoT solutions. These investors typically seek high-growth potential and alignment with sustainable development goals. Additionally, **Strategic Corporate Investment** or partnerships will be pursued with large industrial corporations and conglomerates that have significant salt waste challenges and are looking to adopt advanced waste valorization technologies. Such partnerships can provide not only capital but also access to pilot sites, industry expertise, and established market channels.

3.6.8. Regulatory Considerations

Navigating the complex landscape of regulatory compliance is paramount for the successful deployment and commercialization of the framework. Key regulatory considerations include:

- **Data Privacy Regulations:** Adherence to national and international data privacy laws (e.g., GDPR principles, regional data protection acts) is critical, especially given the handling of potentially sensitive industrial operational data. The privacy-preserving nature of FL is a direct response to these requirements.
- **Environmental Regulations:** Compliance with specific environmental discharge limits and waste management regulations pertinent to salt waste valorization processes will be a core functional requirement. The framework's ability to monitor and optimize processes for compliance will be a key selling point.
- **Industrial Safety Standards:** Ensuring the framework's integration with existing industrial safety protocols and standards to prevent operational hazards.
- **Intellectual Property Laws:** Compliance with national and international IP laws, as detailed in Section 3.6.6, is essential to protect the developed technology and avoid infringement.

Proactive engagement with regulatory bodies and legal experts will ensure that the framework is designed and deployed in full compliance with all applicable laws and standards, facilitating smoother market adoption.

3.7. Anticipated Conclusion

The proposed research aims to develop a transformative framework that integrates Federated Learning, Digital Twin technologies, and decentralized Decision Support Systems for the collaborative, privacy-preserving optimization of industrial salt waste valorization. This multi-modal AI framework is anticipated to yield significant advancements in environmental sustainability and resource recovery. By enabling industries to collectively learn from their distributed data without compromising privacy, the framework will unlock unprecedented opportunities for process optimization, leading to substantial reductions in waste volume, increased recovery of valuable resources, and optimized energy and chemical consumption. The real-time predictive capabilities of Digital Twins, coupled with the actionable insights from decentralized Decision Support Systems, will empower operators to make more informed and proactive decisions, transforming reactive waste management into an intelligent, adaptive, and economically viable process. Ultimately, this research is expected to provide a foundational technology for a truly circular economy in salt-related industries, contributing to a more sustainable and climate-intelligent industrial ecosystem.

3.8. Work Breakdown Structure

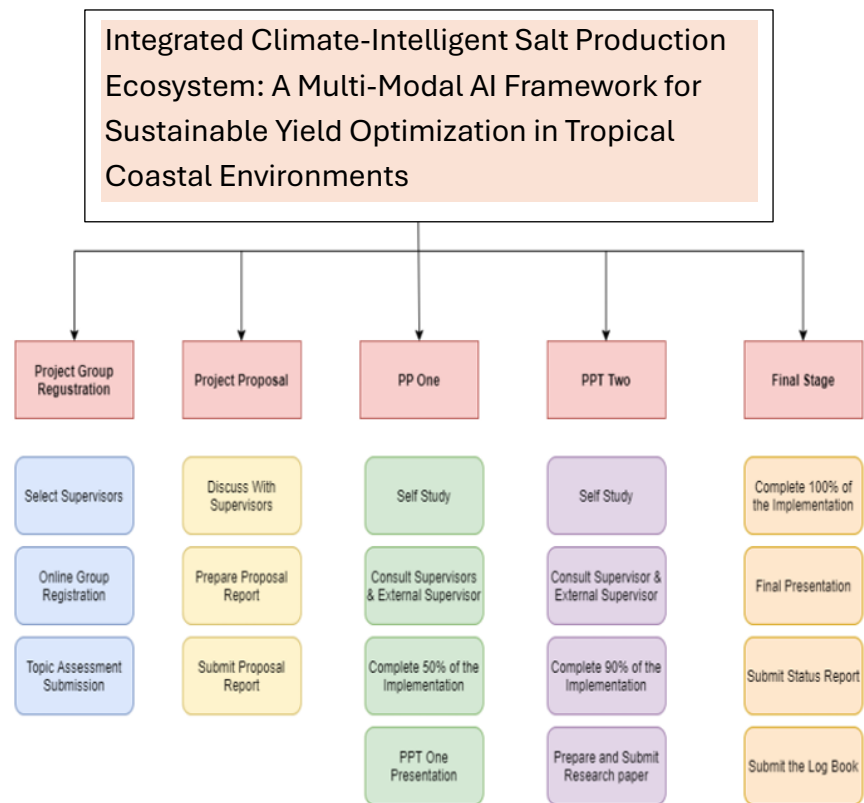


Figure 4 – Work breakdown structure

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

- **Privacy-Preserving Collaborative Learning:** The system must enable multiple industrial entities to collaboratively train a global optimization model for salt waste valorization without requiring direct sharing of raw, sensitive operational data. This core functionality will rely on Federated Learning principles, where only model updates are exchanged.
- **Real-time Process Monitoring and Digital Twin Synchronization:** The framework must continuously collect sensor data (e.g., pH, temperature, conductivity, flow rates, chemical composition, energy consumption) from physical valorization processes. This data must be accurately and in real-time reflected in corresponding digital twin models, providing an up-to-date virtual replica of the physical operations.
- **Predictive Analytics for Anomaly Detection and Performance Forecasting:** The Digital Twins, powered by the collectively trained FL models, must be capable of predicting process anomalies, potential equipment failures, and forecasting key

performance indicators. These KPIs include resource recovery yield, waste reduction metrics, and energy/chemical consumption efficiency.

- **Optimization Recommendations:** The Decision Support System must generate actionable recommendations for optimizing valorization parameters. These recommendations will cover aspects such as optimal chemical dosing, temperature control, flow rates, and treatment times, all aimed at maximizing resource recovery and minimizing waste generation and operational costs.
- **Decentralized Decision Support Interface:** The system must provide intuitive dashboards and interfaces for local operators and managers. These interfaces will allow users to visualize Digital Twin insights, receive DSS recommendations, and interact with the system to implement operational adjustments in a decentralized manner.
- **Traceability and Auditability:** The framework must maintain a verifiable and immutable record of model updates, data provenance, and decision-making processes. This capability is crucial for accountability and compliance, potentially leveraging blockchain integration as suggested for secure FL-DT systems

4.2. User Requirements

4.2.1. Customer Requirements

Industrial customers, typically plant owners, managers, and corporate stakeholders, will prioritize the following:

- **Quantifiable Return on Investment (ROI):** The system must demonstrate clear economic benefits, such as reduced operational costs (e.g., lower energy and chemical consumption), increased revenue from recovered resources, and decreased waste disposal expenses.
- **Environmental Compliance and Sustainability:** The framework must help achieve and maintain compliance with environmental regulations, reduce ecological footprint, and enhance the company's sustainability profile.
- **Data Privacy and Intellectual Property Protection:** Assurance that sensitive operational data and proprietary processes remain confidential and secure, even when participating in collaborative learning initiatives.
- **Reliability and Uptime:** The system must be robust and reliable, ensuring continuous operation of critical valorization processes without introducing new points of failure.
- **Scalability and Adaptability:** The solution should be easily scalable to multiple facilities and adaptable to different salt waste compositions and valorization technologies.

- **Ease of Integration:** The framework should seamlessly integrate with existing industrial control systems (OT) and IT infrastructure with minimal disruption.

4.3. System Requirements

4.3.1. Hardware Requirements

Edge Computing Devices: Industrial PCs or embedded systems at each facility, capable of running local FL clients and DT instances. These require sufficient CPU/GPU processing power, memory (RAM), and storage for local data processing and model training.

4.3.2. Software Requirements

Programming Languages: Python (for AI/ML models), JavaScript (for web dashboard).

Frameworks and Libraries:

- TensorFlow / PyTorch (machine learning and federated learning).
- TensorFlow Federated / Flower (federated learning framework).
- Scikit-learn, NumPy, Pandas (data analysis and preprocessing).
- OpenCV (for image/data processing if needed).

Database Systems: MongoDB or PostgreSQL to store sensor data, logs, and outputs.

Web Development Tools: React (front-end), Node.js + Express (back-end).

Development Tools: Visual Studio Code / PyCharm (coding and debugging).

Version Control: Git and GitHub/GitLab for collaboration and code management.

Simulation Tools: MATLAB or COMSOL (for physics-informed data simulation).

Visualization Tools: Matplotlib, Plotly, or Dash for graphs and dashboards.

Project Management Tools: Jira

4.3.3. Non-Functional Requirements

- **Privacy:** Adherence to strict privacy-by-design principles, ensuring that raw sensitive industrial data never leaves its source.
- **Security:** The system must be resilient against various cyber threats, including data poisoning attacks, inference attacks, and unauthorized access. This necessitates strong authentication, secure communication channels (e.g., TLS/SSL), and robust access control mechanisms.

- **Scalability:** The framework must be capable of scaling horizontally to accommodate a growing number of participating industrial sites and increasing volumes of real-time data without significant performance degradation.
- **Reliability and Robustness:** The system must maintain continuous operation and accurate performance despite potential client dropouts, network intermittency, sensor failures, and data heterogeneity. Fault-tolerant aggregation algorithms and adaptive learning strategies are essential.
- **Maintainability:** The modular architecture and use of standard frameworks should facilitate easy updates, debugging, and long-term maintenance.

4.3.4. Use Cases

- **Real-time Process Optimization:** Adjusting chemical dosing, temperature, and flow rates based on live data to maximize resource recovery and minimize waste generation.
- **Anomaly Detection:** Identifying abnormal patterns in process data that indicate potential malfunctions, contamination, or inefficiencies in real-time.
- **Resource Recovery Yield Forecasting:** Predicting the quantity and quality of valuable minerals (e.g., salts, lithium) that can be recovered from a given input of salt waste.
- **"What-If" Scenario Simulation:** Allowing operators to virtually test the impact of changes in waste input, process parameters, or market conditions on valorization outcomes.

4.3.5. Test Cases

- **FL Model Accuracy & Convergence:** Evaluate global model performance on synthetic and real-world non-IID datasets.
- **Privacy Guarantees:** Verify that differential privacy parameters are met and test against known inference attacks..
- **Fault Tolerance:** Test system behavior under client dropout and network interruptions

4.3.6. Wireframes

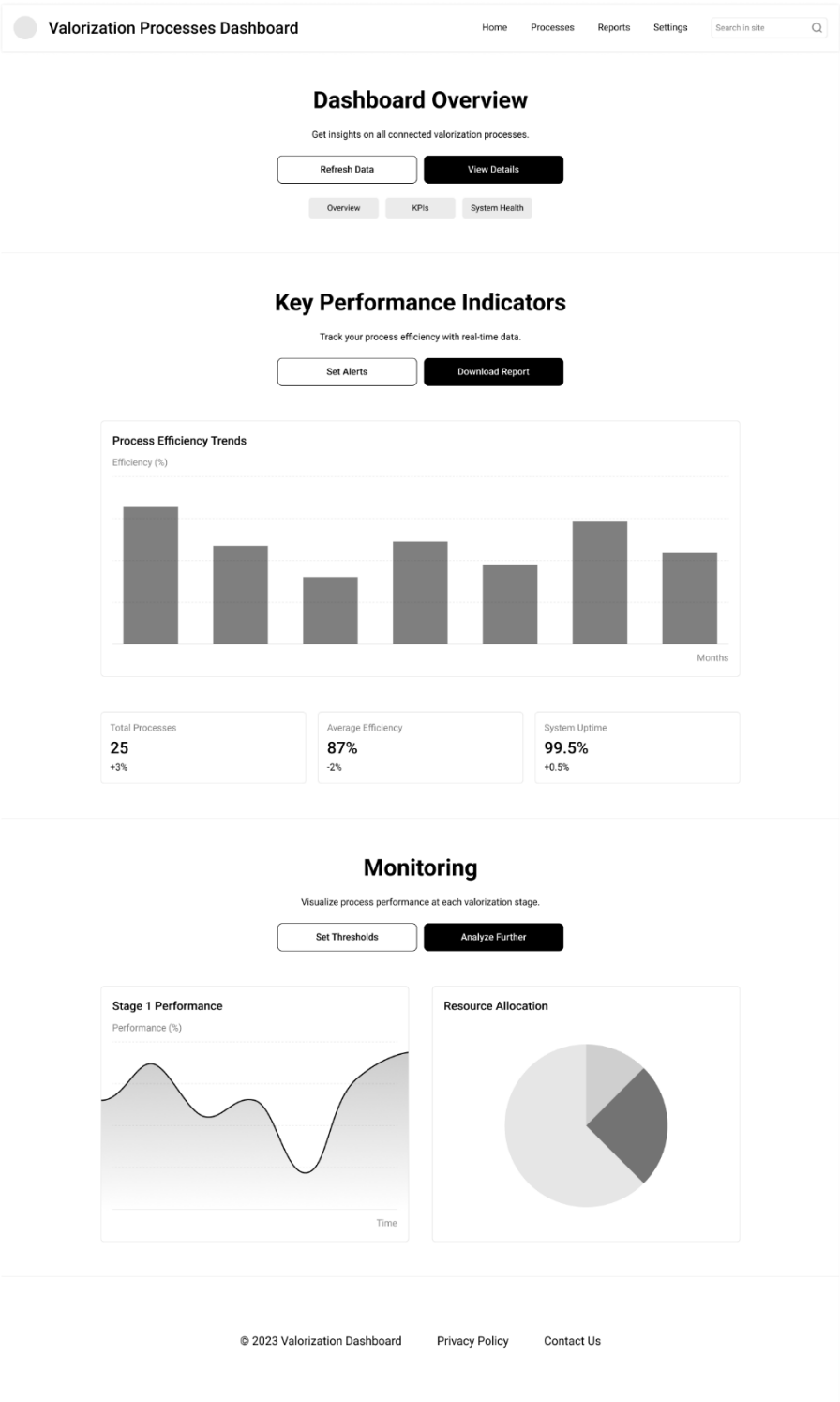


Figure 5 - Wireframe

5. BUDGET AND BUDGET JUSTIFICATION

Expense	Cost (LKR)	Net Total	Justification
Personal Costs			
Travelling	36,000		3000 x 12 covering site visits to Puttalam and supervisor meetings.
Printing	6,000	42,000	500 x 12
Software And Service			
Domain Purchase	1,000		4000 / 4 mem
Host and deploy	17,500		AWS EC2 Instance (200\$) 70,000/4
Perplexity Pro	18,000		6000 x 12 / 4
Premium Api	12,500		For data gathering (e.g.: weather) 50,000 / 4
Google Collab Pro	36,000		For model training (10\$ = 3000 per month)
Cloud Storage	20,000	105,000	Additional storage if needed
Grand Total		147,000	

Table 1 - Budget

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